7 (a) (i)

$T^2 \propto r^3$

$(2\pi r/v)^2 \propto r^3$

$r^2 / v^2 \propto r^3$

$a \propto 1/r^2$ and thus we can conclude F is inversely proportional to r^2 since $F = ma$

(ii)

Gravitational force provides the centripetal force for circular motion to take place.

By Newton's 2nd Law,

$F = ma$

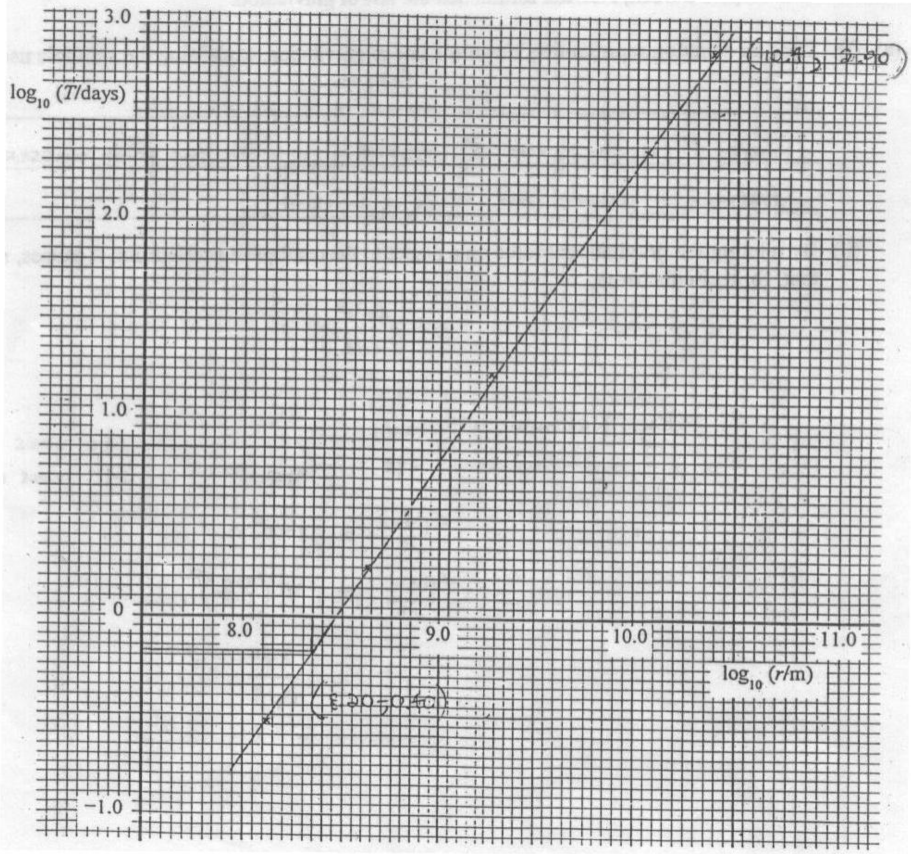
$GMm/r^2 = mr\omega^2$

$GM/r^3 = (2\pi/T)^2$

$T^2 = 4\pi^2 r^3 / GM$

(b) (i)

lg (T/days)	lg (r/m)
2.38	10.05
1.22	9.27
0.25	8.63

		(ii)		
		(iii)	Gradient = 2.90 – (-0.40) / (10.40 – 8.20) = 1.50	
		(iv)	$T^2 = 4\pi^2 r^3/ GM$ $T^2 = kr^3$, where $k = 4\pi^2 / GM$ (constant because all the moons are orbiting around Jupiter) lg T = (3/2) lg r + lg k If lg T is plotted against lg r, the gradient should be 1.5 and this corresponds with the graphically obtained value. Hence, the data support the relation.	
	(c)		T_G = 7.16, lg T_G = 0.85 From graph, lg (r_G) = 9.05, $r_G = 1.12 \times 10^9$ m	
	(d)		T_T = 16.2 / 24 days, lg T_T = -0.17 From graph, lg (r_G) = 8.35, $r_G = 2.24 \times 10^8$ m r_G calculated is the distance of the Thebe to the centre of Jupiter. We can only decide on the accuracy of the statement if the radius of Jupiter is known.	
	(e)		No, because the k value mentioned in (c)(ii) will be different. Orbital radii and periods of moons depend on the mass of the planet where the moons are orbiting.	